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RESEARCH ARTICLE

Genetic structure and diversity of spotted gar (*Lepisosteus oculatus*) at its northern range edge: implications for conservation

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Abstract Understanding the spatial context of genetic variation for species at risk is important for effective management and long-term survival of the species. We use multilocus microsatellite data to investigate the population genetic structure of the spotted gar (*Lepisosteus oculatus*) across its northern range edge in Canada. We then compare these northern individuals with samples taken from the southern core of the species range. For the northern samples, significant genetic differentiation among groups of individuals forming two major genetically distinct populations, and as many as 7–9 smaller subpopulations, was recovered using hierarchical Bayesian assignment methods and non-equilibrial discriminant function analyses. Spatial genetic variation is present, particularly at higher hierarchical groupings; however, some population admixture at sites is evident and is indicative of dispersal and gene flow among some locations or shared ancestry. Gene flow estimates among populations and subpopulations is very low, ranging from essentially complete isolation to as high as 5 %—suggesting that mechanisms in addition to geographic isolation are operating to create genetic structure. In Lake Erie, the physical isolation of Point Pelee appears to confer distinct genetic differentiation for those populations and provide a source of genetic variation for Lake Erie proper when breaches to the barrier beach occur. Results indicate that the northern edge populations are distinct from southern populations and should be conserved to maintain the overall genetic diversity of this species. Additionally, the asymmetrical genetic connectivity among the Point Pelee and Rondeau Bay sites highlights the sensitivity of Point Pelee to environmental perturbation and habitat degradation.

Keywords Spotted gar · Microsatellite · Edge population · Genetic structure

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Introduction

Conservation strategies for species at risk aim to consider range-wide genetic variation and diversity (e.g. Reed and Frankham 2003) and attempt to afford protection not only at the species level, but also to include genetically distinct populations (Jelks et al. 2008). In contrast to core populations, populations at the edge of a species' range may display disproportionately higher genetic variation (Munwes et al. 2010) as edge populations tend to be more isolated, exhibit smaller population sizes, and experience increased levels of genetic drift and differentiation (Garcia-Ramos and Kirkpatrick 1997). The detection of genetic diversity within a species, including the identification of

distinct populations that may be of conservation importance, can be accomplished through the use of contemporary molecular genetic techniques such as genotyping individuals at polymorphic microsatellite loci (e.g. Rodriguez-Munoz et al. 2007). Management actions can then be taken to preserve the range-wide genetic variation.

Spotted gar (Lepisosteidae: *Lepisosteus oculatus*) is native to the Mississippi and Great Lakes drainages of North America, ranging from the Gulf of Mexico northward to the Great Lakes region (Page and Burr 2011). The species is at the northern edge of its range in Canada, where it is listed as *Threatened* in Canada under the Canadian *Species at Risk Act* and the *Endangered Species Act* in the province of Ontario, due to its limited distribution and susceptibility to potential habitat loss (COSEWIC 2005). Within Canada, the spotted gar is found in three coastal wetlands of Lake Erie: Long Point Bay; Point Pelee; and, Rondeau Bay, the Canadian waterbody with the highest reported density of individuals (COSEWIC 2005). In the USA, Spotted gar is assigned *Special Concern* conservation status or higher in all of Great Lakes basin states, but is generally common south of the Great Lakes region where it is not of conservation concern (NatureServe 2014). To date, there has been no comprehensive study of the genetic diversity for spotted gar within the northern portion of its distribution, or the diversity of northern populations relative to that of southern populations. Due to the large area of unsuitable habitat between the northern sample sites likely limiting dispersal and the disjunct nature of the species' range, our a priori expectations are that each sample site in the north will constitute a distinct population and the northern populations will be distinct from the southern populations.

A comprehensive recovery strategy for the species in Canada (Staton et al. 2012) has outlined several urgent research priorities. These priorities include the determination of the degree of connectivity between Canadian populations, determination of population size, and assessing the changes in population characteristics over time. Here, we use multilocus microsatellite data to assess the population genetic variation of spotted gar within the Canadian portion of its distribution to determine: (1) the number of genetically distinct populations and their effective number of breeding individuals (N_b), a measure related to effective population size (see Waples 2005); (2) the presence of spatial structure among populations; (3) the genetic connectivity, if any, among populations/habitats; and, (4) the presence of structure among year classes in the Rondeau Bay population to determine if the population structure is changing over time. The information gained through this study will address these urgent research priorities for spotted gar in Canada.

Methods

Specimens were collected by electrofishing, fyke nets, and gill nets in Canada and the USA, with non-lethal fin tissue samples taken and preserved in 95 % ethanol (see Table 1). Additional tissue samples were provided by researchers from Tulane University and Mississippi State University (see Acknowledgements section). Study-wide, fin tissue samples were obtained from 681 specimens collected from eight different sampling locations (Table 1; Fig. 1). Genomic DNA was extracted following the plate-based protocol of Elphinstone et al. (2003).

Eight microsatellite loci (Moyer et al. 2009) developed for Alligator Gar (*Atractosteus spatula*) were amplified for all sampled individuals using polymerase chain reaction (PCR). PCR reactions were conducted in 12.5 μL reactions containing 0.5 M IR-dye labelled forward primer, 0.5 M reverse primer, 1× PCR buffer (Applied Biosystems), 200 μM of each dNTP, 0.5 U Taq polymerase and various volumes of 25 mM $MgCl_2$ (Table 2). The thermal cycler reaction conditions were as follows: initial denaturation at 94 °C for 120 s, followed by 35 cycles of 94 °C for 15 s, various annealing temperatures (Table 2) for 15 s, 72 °C for 30 s, and a final extension period of 60 s at 72 °C.

Dye-labelled PCR products were visualized using a LiCor 4300 DNA analyzer (Li-Cor Biosciences Inc.) with manufacturers' size standards (50–350 bp). Individual genotypes were determined by scoring alleles using GENE IMAGIR v.4.05 (Scanalytics Inc.). Genotype data were tested for allele scoring error, outlier alleles, discrepancies between observed and expected allele step sizes, and large allele gaps in MSA v4.05 (Dieringer and Schlötterer 2003).

Tests for linkage disequilibrium (LD), adherence to Hardy–Weinberg Equilibrium (HWE), and pairwise F_{ST} (Weir and Cockerham 1984) were performed on all sites and recovered populations in ARLEQUIN v.3.15 (Excoffier et al. 2005) followed by sequential Bonferroni correction (Rice 1989). Recovered populations were further characterized by calculating observed (H_0) and expected (H_E) heterozygosities and allelic richness (A) for each site and population in ARLEQUIN. BOTTLENECK v.1.2.02 was used to test for indicators of recent bottleneck effects for both sites and populations, employing a Wilcoxon signed-rank test (TPM model). Effective number of breeding individuals for Point Pelee and Rondeau Bay in the north, and for populations in the south were estimated in LDNE (Waples and Do 2008) using 0.02 as the lowest allele frequency to provide a balance between precision and bias (Waples and Do 2010).

To infer genetic structure and the number of populations, we used STRUCTURE v.2.3 (Pritchard et al. 2000) with location priors in 5 replicate runs. Admixture was

Table 1 Location and number of spotted gar sampled from each site

Site	Site Name	Waterbody	Latitude	Longitude	No. of samples
LA	Louisiana	Lac des Allemands	29.9151°N	90.5728°W	39
MS1	Mississippi 1	Pascagoula River	30.7749°N	88.6888°W	10
MS2	Mississippi 2	Townsend Lake	33.1439°N	90.4110°W	38
MO	Missouri	Mingo Creek	36.97966°N	90.2047°W	42
MI	Michigan	Loon Lake/Lake Pleasant	41.8689°N	84.9427°W	8
PP	Point Pelee	Lake Erie	41.9655°N	82.5094°W	92
RB	Rondeau Bay	Lake Erie	42.2873°N	81.8978°W	250
LPB	Long Point Bay	Lake Erie	42.6145°N	80.4503°W	5

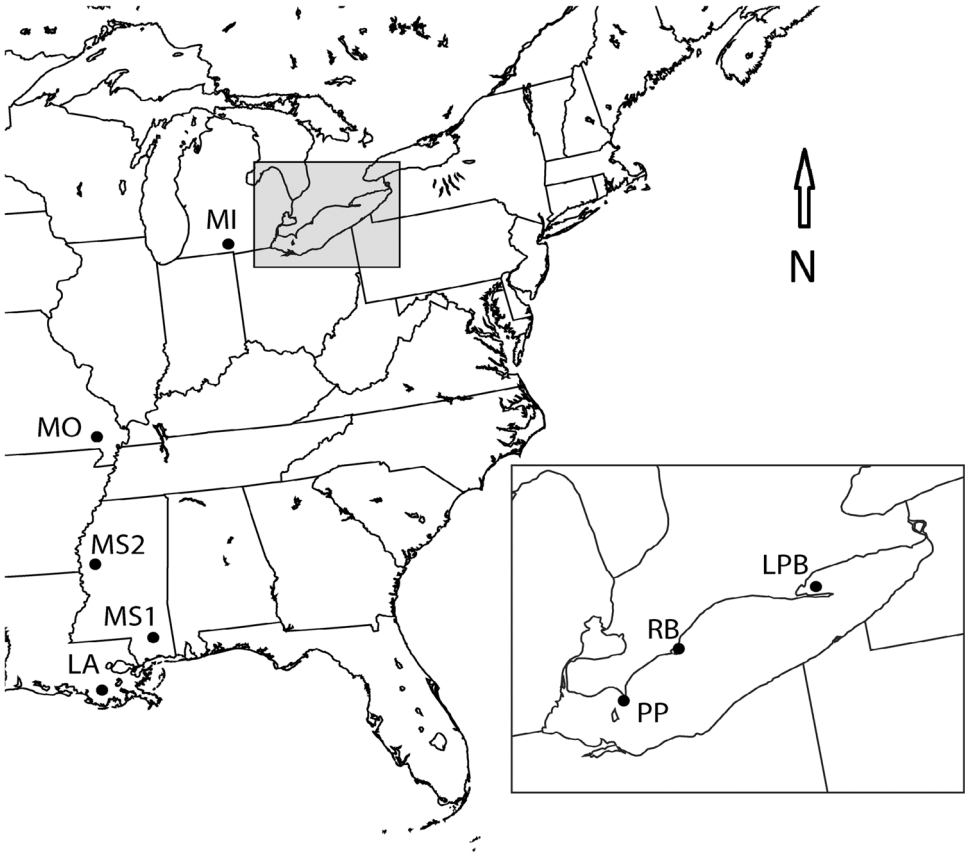

Fig. 1 Spotted gar sample locations, indicated by *black dots* on the map (see Table 1 for key to labels). Inset is an enlarged view of the shaded region

allowed and the simulation was run for K = 1–10, using a 100,000 burn-in period and 1,000,000 Monte Carlo Markov Chain generations. The Evanno et al. (2005) ΔK method was used to select the number of clusters in STRUCTURE HARVESTER (Dent and vonHoldt 2012); replicate runs were compiled in CLUMPP v.1.1.2 (Jakobsson and Rosenberg 2007). Individuals in the dataset were reassigned to a specific group according to Q-membership values, and the STRUCTURE-ΔK procedure was repeated for each recovered group to search for additional substructure. The process was repeated until no remaining structure could be detected (K = 1). Outputs from the STRUCTURE runs were visualized using DISTRUCT v.1.1 (Rosenberg 2004). For comparative purposes, we used the *k*-means clustering algorithm in DAPC (Jombart et al. 2010) to decompose individuals into smaller genetic groups and compared these results with those from STRUCTURE. Simulations were run with locational priors included (loc) and again without locational priors included in the analysis (no loc).

Table 2 PCR forward primers and reaction conditions for amplification of microsatellite loci. Primers were originally developed by Moyer et al. (2009)

Primer	$MgCl_2$ (μL)	Ta(°C)	Number of alleles	Size range (bp)
Asp010	1.0	58.3	4	287–293
Asp012	1.0	60.2	5	206–218
Asp029	1.0	58.3	5	116–124
Asp046	1.1	60.2	9	183–199
Asp057	1.0	60.2	14	148–187
Asp066	1.0	62.0	18	225–279
Asp095	1.0	63.5	14	182–224
Asp096	0.75	58.3	7	104–114

Connectivity (i.e., contemporary gene flow) was estimated in BAYESASS v.3.0 (Wilson and Rannala 2003) and BIMr (Faubet and Gaggiotti 2008) at the initial breaks in the hierarchy (e.g., northern 2 groups, blue and yellow see Fig. 2), and then for all final recovered 'populations'. BAYESASS runs were performed using 10^7 burn-in, followed by 10^8 sampling iterations, and all run traces were evaluated for MCMC convergence in TRACER v1.6 (Rambaut and Drummond 2007). Asymmetrical gene flow among Point Pelee and Rondeau samples was also independently tested using BIMr (Faubet and Gaggiotti 2008) in triplicate runs using 10^5 burn-in and sample size, and other parameters set to default values.

To examine structure among year classes in Rondeau Bay, pectoral fin rays were collected from 273 individuals captured at this location. These pectoral fin rays were used to age specimens in the manner of Glass et al. (2011) and determine the natal year for each individual. All of these individuals were genotyped and included in the population-level analyses described above. Once the age of each individual was identified, we assigned individuals to groups based on their natal year and population assignment from previous STRUCTURE analysis. We also reran STRUCTURE on all samples across year classes (without location information). HWE departures were calculated based on the individual assignments to populations within Rondeau Bay using the same procedures as above.

Results

Microsatellite genotypes were recovered for a dataset totalling 484 individual fish. All microsatellite loci were polymorphic with the number of alleles ranging from 5 to 19 (Table 2). The allelic richness (A) ranged from 1 to 15 alleles at a single locus per site (Table 3) with observed (H_0) and expected (H_E) heterozygosities ranging from 0.00 to 1.00 and 0.115 to 0.942, respectively. The tests for population bottlenecks in the northern samples did not indicate mode-shifts for any of the populations, but did show significant heterozygosity excess for the Point Pelee population (two-tailed; $P = 0.00391$), for Pop 3 (one-tailed, $P = 0.03711$), and for Pop 4 (one-tailed, $P = 0.03711$) from the Rondeau Bay-Michigan-Long Point Bay group.

Following sequential Bonferroni correction, 15 of the 64 possible tests showed significant deviation from Hardy–Weinberg equilibrium and three additional loci were monomorphic within the Michigan site, likely due to small sample size ($n = 10$). For the southern samples, each of the sites had a single locus that deviated from HWE, although the loci in question differed among each locale. For the northern sample sites, both Point Pelee (5 of 8 loci) and Rondeau Bay (7 of 8 loci) deviated significantly from Hardy–Weinberg equilibrium. When the individuals from the Rondeau Bay, Michigan, and Long Point sites were assigned to their STRUCTURE populations, 13/40 deviations from HWE persisted following Bonferroni correction (see supplementary Table S1).

Pairwise F_{ST} values among sampling sites ranged from −0.010 to 0.297 (Table 4). The pairwise F_{ST} comparison between the Missouri and Mississippi (Townsend Lake) populations showed no significant divergence, while the Long Point individuals were not significantly differentiated from the populations in Rondeau Bay, Missouri, and both Mississippi sites. All other comparisons showed significant divergence.

The hierarchical search for the number of populations in STRUCTURE yielded eight populations when locational priors were included in the simulation (loc) and twelve populations when locational priors were excluded (no loc) that could not be further decomposed into smaller population groups—two (loc) to three (no loc) populations from the southern samples, and up to eight (loc) and nine (loc) in the northern range. The first break in the hierarchy split the southern USA samples into a single group (red in Fig. 2), and two groups in the north (yellow and blue in Fig. 2). Population admixture was evident at all sampling sites across the hierarchy, but was minimal at Point Pelee at the $K = 2$ hierarchy (yellow in Fig. 2). Further STRUCTURE analysis of the northern samples using location priors produced nearly identical results at the first hierarchy, indicating the importance of location in the clustering of Point Pelee samples from other northern samples (Supplementary Fig. S1). The Rondeau Bay-Michigan-Long Point Bay group (light blue in Fig. 2) was further divided into five populations. Estimates of N_b and connectivity were then taken among these six resolved populations from the northern region.

Without the use of location data, the northern group five (dark blue in Supplementary Fig. S1) was decomposed into

Table 3 Allelic richness (a), observed heterozygosity (H_0) and expected heterozygosity (H_E) for eight microsatellite loci at each sample site

Sample site	Asp010			Asp012		
	A	H_0	H_E	A	H_0	H_E
Louisiana	4	0.67568	0.63532	3	0.16216	0.19993
Mississippi 1	3	0.90000	0.67368	3	0.40000	0.35263
Mississippi 2	3	0.60000	0.66045	2	0.13514	0.17364
Missouri	3	0.57895	0.62421	3	0.11905	0.11503
Michigan	2	0.62500	0.45833	1	0	0
Pt. Pelee	4	0.69767	0.61954	2	0.35870	0.29597
Rondeau	4	0.54505	0.56512	5	0.13362	0.13932
Long Pt	3	0.60000	0.51111	3	0.25000	0.60714
Sample site	Asp029			Asp046		
	A	H_0	H_E	A	H_0	H_E
Louisiana	3	0.34375	0.37775	4	0.62857	0.71677
Mississippi 1	3	0.30000	0.27895	4	0.55556	0.69935
Mississippi 2	2	0.46154	0.40121	5	0.69444	0.73279
Missouri	3	0.69768	0.49611	5	0.67568	0.77046
Michigan	1	0	0	3	0.25000	0.49167
Pt. Pelee	3	0.30769	0.47816	4	0.39326	0.64680
Rondeau	5	0.22105	0.40269	8	0.51037	0.72869
Long Pt	2	0.40000	0.35556	4	0.40000	0.71111
Sample site	Asp057			Asp066		
	A	H_0	H_E	A	H_0	H_E
Louisiana	12	0.86842	0.84632	14	0.84211	0.88702
Mississippi 1	5	0.66667	0.83007	12	0.90000	0.94211
Mississippi 2	10	0.85714	0.83106	12	0.91892	0.89856
Missouri	9	0.73810	0.85055	14	0.88095	0.90734
Michigan	1	0	0	5	0.62500	0.68333
Pt. Pelee	6	0.56044	0.64975	5	0.65934	0.69455
Rondeau	9	0.24359	0.38734	10	0.70886	0.79400
Long Pt	4	0.20000	0.73333	6	1.0	0.86667
Sample Site	Asp095			Asp096		
	A	H_0	H_E	A	H_0	H_E
Louisiana	12	0.76923	0.82950	3	0.15152	0.33986
Mississippi 1	7	1.00000	0.86316	4	0.14286	0.71429
Mississippi 2	10	0.72727	0.84522	3	0.25000	0.43205
Missouri	10	0.72727	0.88671	3	0.25000	0.42180
Michigan	4	0.75000	0.69167	2	0.40000	0.35556
Pt. Pelee	4	0.54237	0.61495	4	0.40000	0.59213
Rondeau	9	0.46836	0.73547	5	0.16265	0.73308
Long Pt	2	0.25000	0.535714	2	0.33333	0.33333

an additional three populations; however, both likelihood values for K = 1, K = 2, K = 3 were similar and Q-values for individuals showed evidence of individual admixture among the three groups (see Supplementary Fig. S1).

Measures of contemporary connectivity varied given the individual clustering method used prior to BAYESASS. For the location-based groupings in the northern samples, BAYESASS indicated that asymmetrical gene flow among

Table 4 (a) Pairwise F_{ST} comparisons among all pairs of spotted gar sample sites, (b) Pairwise F_{ST} comparisons among all STRUCTURE population pairs from the Northern Range. Values in bold indicate a significant difference ($P < 0.005$)

Site	LA	MS1	MS2	MO	MI	Pt. Pelee	Rondeau	Long Pt.
Sample sites comparisons								
LA	0							
MS1	**0.0813**	0						
MS2	**0.0541**	**0.0774**	0					
MO	**0.0437**	**0.0706**	0.0043	0				
MI	**0.2254**	**0.1326**	**0.2166**	**0.1923**	0			
Pt. Pelee	**0.2425**	**0.1994**	**0.2004**	**0.1894**	**0.2774**	0		
Rondeau	**0.1668**	**0.0899**	**0.1455**	**0.1238**	**0.0785**	**0.1202**	0	
Long Pt.	**0.1801**	0.0732	0.1123	0.0968	**0.2451**	**0.2369**	0.1035	0

Pop	1	2	3	4	5	6
Northern Range STRUCTURE populations						
1	0					
2	**0.1872**	0				
3	**0.2005**	**0.1688**	0			
4	**0.1376**	**0.1523**	**0.2340**	0		
5	**0.0406**	**0.1436**	**0.1895**	**0.1427**	0	
6	**0.1418**	**0.0935**	**0.1451**	**0.1678**	**0.1264**	0

Point Pelee and Rondeau Bay with an $m_{RB \rightarrow PP}$ of 0.0082 (SD = 0.0073), and an $m_{PP \rightarrow RB}$ of 0.0446 (SD = 0.0116). From the individual assignments, no individuals caught in Point Pelee were determined to be migrants; however, 37 individuals caught at Rondeau Bay had posterior probability estimates of migrant ancestry (1st or 2nd generation, no emigrants detected) >0.10. Gene flow estimates were largely the same when taken among the six resolved populations, with asymmetrical gene flow from Point Pelee (Pop 6) into Rondeau Bay (Pop 4). The Point Pelee (Pop 6) showed a comparatively low N_b estimate compared to other recovered populations, except Long Point Bay (Pop 3). Pop 3 showed the lowest N_b compared to all other recovered populations and also appeared to be a sink population, receiving migrants from each of the other five. The BIMr migration estimates also strongly indicated asymmetrical gene flow from Point Pelee into Rondeau Bay; $m_{PP \rightarrow RB}$ of 0.1133 (SD = 0.0270) and $m_{RB \rightarrow PP}$ of 0.0199 (SD = 0.0181).

For the STRUCTURE runs without location information, m ranged from 0.0059 to 0.1222, with the highest migration among sites within the Point Pelee-Rondeau Group (groups 2 and 3, Fig. 2). Estimates of m using the nine recovered populations (Fig. S1) yielded lower self m rates and receiving m rates greater than 0.01 from all sites for populations 5, 6, 7 (Supplementary Fig. S1) suggesting that these groupings may be problematic, further supporting the decision to consider groups 5, 6, and 7 as a single population unit.

Specimens from Rondeau Bay ranged from 2 to 10 years in age, and their corresponding natal year ranged from 1998 to 2007. Three STRUCTURE populations were recovered among the aged individuals in Rondeau, with admixture evident across years—indicative of no structure among year classes.

Discussion

Significant genetic structure was revealed among spotted gar sampled throughout their Canadian range and comparative southern samples. Within the north, a total of eight genetic 'populations' were recovered using Bayesian assignment methods, with considerable evidence for genetic isolation and a population bottleneck for the Point Pelee samples. This finding is consistent with the physical isolation of the Point Pelee marsh from the other sampling sites within Lake Erie. Furthermore, estimates of contemporary gene flow indicate that the Point Pelee population acts as an exporter of individuals to Lake Erie. An additional five potential populations were recovered within the northern range, each highly differentiated and exhibiting varying, but in general, very low levels of genetic connectivity among each other. These results are contrary to our a priori expectation that each northern sample site will constitute a separate, distinct population and indicate that other factors in addition to physical isolation play a role in the genetic structuring of spotted gar populations in the northern range.

The large geographical pattern of genetic structure between the northern-edge populations and the southern core population (Fig. 2) is consistent with postglacial dispersal

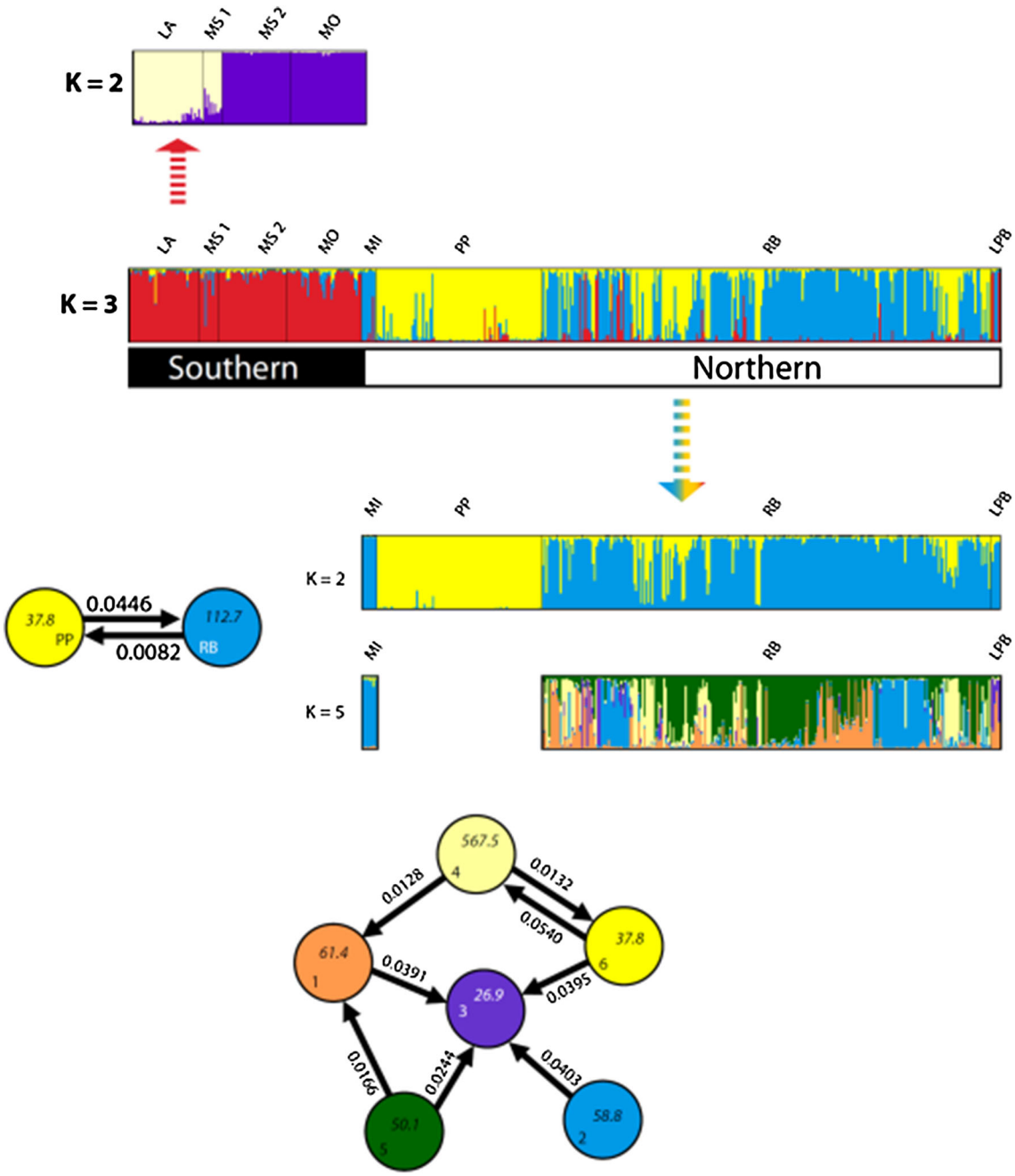

Fig. 2 Population genetic structure of spotted gar determined by Bayesian clustering assignment using STRUCTURE, using location information showing eight recovered populations. Southern group population substructure for K = 2, and northern group substructure shows the isolation of the Point Pelee population, and the admixture within the remaining Northern sites. The *coloured bubbles* correspond to the respective recovered population, with effective number of breeding individuals (N_b) in *italics*; *arrows* indicate direction of gene flow with accompanying migration estimates

and subsequent isolation (Garcia-Ramos and Kirkpatrick 1997) and supports our a priori expectation that northern populations would be distinct from southern populations. Other studies relying on mtDNA sequence data (Glass 2012) demonstrated the presence of shared haplotypes across the sampled range of spotted gar, indicating that microsatellite divergence is likely a product of post glacial isolation.

In the northern range, genetic admixture across sampling sites was evident for all sites except Point Pelee. Individuals sampled from Michigan (Loon Lake/Lake Pleasant) and Long Point Bay (Lake Erie) displayed genotypes consistent with the groups recovered from Rondeau Bay (Lake Erie). For the Long Point Bay samples, within-lake dispersal and subsequent gene flow is possible

and, therefore, a likely source of the observed genetic admixture. The observed admixture of the inland Michigan samples with Lake Erie samples indicates a shared ancestry due to postglacial dispersal through glacial Lake Maumee in the present-day Erie basin (Bailey and Smith 1981; Larson and Schaetzl 2001).

The presence of five sympatric populations in the Rondeau Bay site is surprising, especially given the high degree of differentiation among them. What remains unknown is the mechanism(s) that drives and maintains genetic structure among individuals in Rondeau Bay. Tracking data show no evidence of discrete groups of Spotted Gar in Rondeau Bay throughout the summer period (Glass et al. 2012), thus, individuals are not segregated based on population throughout the entire year. One possible explanation for the maintenance of this genetic structure is spawning-site philopatry, where individuals from several populations may intermingle throughout much of the year and return to spawn in areas specific to their population. Spawning-site philopatry has been shown to maintain genetic structure between Walleye (*Sander vitreus*) stocks within a single waterbody (Stepien and Faber 1998). Homing behaviour, which has not previously been demonstrated in gar species, is an avenue of future research that should be pursued through long-term tracking studies to determine if individuals return to the same spawning location in multiple years.

The lack of the genetic differentiation between year classes at the Rondeau Bay site is expected as this species is long lived (Ferrara 2001; Glass et al. 2011) and will spawn in multiple years once sexual maturity is reached (Redmond 1964), with generational overlap maintaining the genetic structure over time. Census population size for Rondeau Bay is estimated in excess of 8000 individuals (Glass et al. 2012); whereas, effective number of breeders is 113.8 when considered as whole, and as high as 567.5 in Pop 4. This large census population size, along with relatively large effective number of breeders (Pop 4) suggests some degree of buffer against drift effects and a viable population of Spotted Gar at this location provided that sufficient habitat is protected.

Probably the most significant finding concerns the isolation of the Point Pelee marsh population from other Lake Erie sites. Both the hierarchical genetic structure and asymmetrical gene flow estimates among the Point Pelee population and others in the Rondeau Bay-Michigan-Long Point Bay group indicate the dispersal of individuals, and the export of alleles from the Point Pelee population. The approximately 5–11 % per generation migration rate out of Point Pelee into Rondeau Bay, coupled with the presence of private alleles, means that Point Pelee may act as a source of genetic diversity for the lake, with the physical isolation of the Point Pelee marsh limiting immigration, increasing drift effects, and promoting the genetic divergence of the marsh population from those in the lake proper. The comparatively low N_b estimates and evidence of genetic bottlenecks further lend support to the hypothesis of Point Pelee acting as a source of diversity for Lake Erie. The export of alleles from Point Pelee via dispersing/emigrating individuals is facilitated by water breaches of the barrier beach at Point Pelee. Periodic breaches have been recorded in the past, with at least seven reported since 1973, and another eight breaches that have been predicted to have occurred (Surette 2006). These breaches likely result in emigration only as Point Pelee individuals have a much greater chance of using the breach due to close proximity and attractant flow. The area between Point Pelee and Rondeau Bay largely lacks shallow, weedy habitat suitable for spotted gar (Staton et al. 2012). This lack of suitable habitat, coupled with the infrequent nature of breach events, likely limits immigration into the Point Pelee marsh from individuals born elsewhere. Real-time radio-tracking of spotted gar in Rondeau Bay throughout spring and summer months showed no individuals dispersing from the bay for the open water of Lake Erie (Glass et al. 2012), further suggesting that adults likely exhibit at least some degree of spatial residency.

Long Point Bay, the location with the smallest number of spotted gar in Canada (Bouvier and Mandrak 2010), appears to be a sink population. Several of the individuals sampled from Long Point Bay were partially or fully assigned to Pop 3, the population that received migrants from each of the other populations in the analysis. Although there is limited genetic evidence based on the small number of individuals sampled from Long Point Bay, traditional sampling also supports the finding of Long Point Bay as a sink. Sampling has resulted in the fewest number of captures of adult spotted gar of the three Canadian locations where the species is known to persist, despite comparable sampling effort. More recently, targeted sampling for juveniles in Long Point Bay resulted in no captures, while sampling in Rondeau Bay with similar gear, effort, and timing resulted in multiple captures (Glass and Mandrak 2014). The small number of individuals present and the status of the population as a sink indicate that the Long Point Bay population of spotted gar may not be viable in the long term.

Conservation implications

Our findings clearly demonstrate that the northern edge populations of the spotted gar are genetically distinct from those from southern region of the species' range. This highlights the need to protect the northern edge populations of the species to maintain the overall genetic diversity in the species. Furthermore, the asymmetrical, 'source-sink'

genetic connectivity between the two main spotted gar habitats in the northern range highlights the sensitivity of the one specific habitat (Point Pelee) to environmental perturbation and habitat degradation. The current genetic data indicate little to no immigration to the Point Pelee habitat and, thus, should this population become severely impacted or extirpated, it is not likely that it would arise again through natural processes alone. This lack of a 'rescue effect' weighs heavily in consideration for species conservation priority listings in Canada (COSEWIC 2010), further strengthening the argument to maintain protection of the Canadian populations of this species.

Due to the number of recovered populations found in Rondeau Bay, the effective number of breeders for each of these populations is much lower than the total number of individuals inhabiting the bay. The estimated population size in Rondeau Bay is 8121 individuals (Glass et al. 2012). On average, the size of each of the five populations is approximately 1624 individuals, which is larger than the minimum viable population size (MVP) of 712 adults identified by (Young and Koops 2010); however, the populations are not likely to be equal in size. The effective number of breeders for Pop 4 is much larger ($N_b = 567.5$) than that of the other populations found in Rondeau Bay ($26.9 \leq N_b \leq 61.4$), thus, the census size of Pop 4 is also likely much larger than the average, while the census size of the other populations is likely much smaller, potentially below the MVP. Additionally, the N_b of the Point Pelee population (Pop 6) and Pop 3 of the Rondeau Bay-Michigan-Long Point Bay group are lower than 50, leaving these populations vulnerable to immediate negative effects due to inbreeding (Franklin 1980). Each of the populations except Pop 4 has a N_b below 500, which is required for long-term viability and maintenance of genetic variation (Franklin 1980).

While the Point Pelee site is afforded some protection due to its location within a national park, the other Canadian populations are only partially within protected areas. Both the Long Point Bay and Rondeau Bay populations occur within and outside of protected areas (Long Point Provincial Park and Big Creek National Wildlife Area for the Long Point Bay population; Rondeau Provincial Park for the Rondeau Bay population), thus, the habitat outside of the protected areas utilized by these populations is subject to degradation. Additional threats have been identified for the Rondeau Bay and Long Point Bay populations, including habitat modification, vegetation removal, nutrient loading, and turbidity loading from terrestrial sources (Bouvier and Mandrak 2010; Staton et al. 2012). Critical habitats have been identified in Rondeau Bay (Staton et al. 2012) and protection of this habitat, including cessation of vegetation removal, is likely necessary for the persistence of genetic diversity in this region. Spotted gar has been found to utilize areas outside of the designated critical habitat, including agricultural drains tributary to Rondeau Bay (Glass and Mandrak 2014), leaving these populations vulnerable. The protection of each population of spotted gar in Rondeau Bay is vital to maintain the genetic diversity found in the northern edge populations because Rondeau Bay houses a large amount of the species' genetic diversity and the genetic structure at this location is likely to be stable over time. One of these populations (Pop 4) is likely large enough to be viable long term, provided sufficient habitat is protected, although measures to protect, and enhance some of the smaller populations found in Rondeau Bay may be required. The maintenance of a large amount of diversity within the small geographic area of Rondeau Bay appears to be unique among the areas sampled.

Recommendations for conservation

Several actions can be undertaken to ensure the maintenance of the overall genetic diversity in the Canadian populations of spotted gar. For Point Pelee, monitoring of the population size and potential loss of diversity due to inbreeding effects is advised. If population size or diversity declines, supplementing the population with individuals translocated from Rondeau Bay should be considered. For Rondeau Bay, identification of the mechanism(s) sustaining multiple populations is important. If spawning-site philopatry is a found to be a cause, the protection of each of the populations' spawning sites will be important to maintain this diversity. Enhancement of the spawning habitat used by the smaller Rondeau Bay populations should be implemented to facilitate growth of these small populations, and ensure their long-term survival. As a precautionary measure, all currently available spawning habitat should be protected, including the habitat that occurs in agricultural drains. Additionally, natural migration should be facilitated by enhancing nearshore wetland habitat in the areas between Point Pelee and Rondeau Bay, and between Rondeau Bay and Long Point Bay to provide suitable stepping stone habitat for migrating individuals. The improvement of spawning habitat in Long Point Bay is also recommended to increase the level of successful reproduction.

The population genetic structure in the northern edge populations of spotted gar reinforces the need to protect these populations and the habitats that they rely on for survival. The prospects for long-term survival of these populations may be enhanced through timely conservation actions, including the protection and enhancement of nearshore coastal wetland habitat in Lake Erie. Habitat protection and enhancement initiatives should be focussed on the areas in which populations of spotted gar currently

occur, as well as sites between these areas to facilitate migration and gene flow.

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